

AnnotationHub

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Dependencies

This document has the following dependencies:

```
library(AnnotationHub)
```

Use the following commands to install these packages in R.

```
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("AnnotationHub"))
```

Corrections

Improvements and corrections to this document can be submitted on its [GitHub](#) in its [repository](#).

Overview

Annotation information is extremely important for putting your data into context. There are many online resources for doing this, and many different databases organizes different information suing different approaches.

There are multiple ways to access annotation information in Bioconductor.

Here we discuss a new way of doing so, through the package [*AnnotationHub*](#). This package provides access to a ton of online resources through a unified interface. However, each data resource has its own peculiarities, so a user still needs to understand what the different datasets are.

In a recent paper I was involved in (Hansen et al. 2014), I used [*AnnotationHub*](#) to interrogate my data against all transcription factor data available through the ENCODE project. I managed to write the code and conduct the analysis in the matter of a single evening, which I think is pretty awesome.

Other Resources

- The [*AnnotationHub*](#) vignette from the [AnnotationHub webpage](#).
- The [Annotation Resources](#) workflow on Bioconductor contains material on [*AnnotationHub*](#).
- The BioC 2015 conference had a tutorial on [*AnnotationHub*](#); material is available through the [course materials](#) site.

Usage

First we create an `AnnotationHub` instance. The first time you do this, it will create a local cache on your system, so that repeat queries for the same information (even in different R sessions) will be very fast.

```
ah <- AnnotationHub()  
ah
```

As you can see, `ah` contains tons of information. The information content is constantly changing, which is why there is a `snapshotDate`. While the object is big, it actually only contains pointers to online information. Actually downloading all the resources available in an `AnnotationHub` is prohibitive.

The object is organized as a vector, with single-dimension indexing. You can get information about a single resource by indexing with a single `[]`; using two brackets `[[ ]]` downloads the object:

```
ah[1]
```

```
## AnnotationHub with 1 record
## # snapshotDate(): 2015-08-17
## # names(): AH2
## # $dataprovder: Ensembl
## # $species: Ailuropoda melanoleuca
## # $rdataclass: FaFile
## # $title: Ailuropoda_melanoleuca.ailMel1.69.dna.toplevel.fa
## # $description: FASTA DNA sequence for Ailuropoda melanoleuca
## # $taxonomyid: 9646
## # $genome: ailMel1
## # $sourcetype: FASTA
## # $sourceurl: ftp://ftp.ensembl.org/pub/release-69/fasta/ailuopoda_m
## # $sourcelastmodifieddate: 2012-10-12
## # $sourcesize: 693412448
## # $tags: FASTA, ensembl, sequence
## # retrieve record with 'object[["AH2"]]'
```

The way you use [AnnotationHub](#) is by using various tools to narrow down your hub to a single or a small number of datasets. Then you download these datasets for your own usage.

Let us first explore some high-level features of the hub:

```
unique(ah$dataprovder)
```

```
## [1] "Ensembl"
## [2] "EncodeDCC"
## [3] "UCSC"
## [4] "Inparanoid8"
## [5] "NCBI"
## [6] "NHLBI"
## [7] "ChEA"
## [8] "Pazar"
## [9] "NIH Pathway Interaction Database"
## [10] "RefNet"
## [11] "Haemcode"
## [12] "GEO"
## [13] "BroadInstitute"
## [14] "ftp://ftp.ncbi.nlm.nih.gov/gene/DATA/"
## [15] "dbSNP"
```

```
unique(ah$rdataclass)
```

```
## [1] "FaFile"          "GRanges"          "Inparanoid8Db"  
## [4] "OrgDb"           "TwoBitFile"       "ChainFile"  
## [7] "SQLiteConnection" "data.frame"       "biopax"  
## [10] "BigwigFile"      "ExpressionSet"    "VcfFile"
```

(we will discuss many of these data classes in future sessions).

You can narrow down the hub by using one (or more) of the following strategies

- Use `subset` (or `[]`) to do a specific subsetting operation.
- Use `query` to do a command-line search over the metadata of the hub.
- Use `display` to get a Shiny interface in a browser, so you can browse the object.

It is often useful to start with a very rough subsetting, for example to data from a specific species. The `subset` function is useful for doing a standard R subsetting (the function also works on `data.frames`).

```
ah <- subset(ah, species == "Homo sapiens")  
ah
```

```
## AnnotationHub with 24236 records
## # snapshotDate(): 2015-08-17
## # $dataProvider: BroadInstitute, UCSC, Ensembl, dbSNP, NIH Pathway In
## # $species: Homo sapiens
## # $rdaclass: GRanges, BigwigFile, ChainFile, FaFile, VcfFile, biopa
## # additional mcols(): taxonomyid, genome, description, tags,
## #   sourceurl, sourcetype
## # retrieve records with, e.g., 'object[["AH133"]]'
##
##           title
## AH133      | Homo_sapiens.GRCh37.69.cdna.all.fa
## AH134      | Homo_sapiens.GRCh37.69.dna.toplevel.fa
## AH135      | Homo_sapiens.GRCh37.69.dna_rm.toplevel.fa
## AH136      | Homo_sapiens.GRCh37.69.dna_sm.toplevel.fa
## AH137      | Homo_sapiens.GRCh37.69.ncrna.fa
## ...       | ...
## AH49184    | Homo_sapiens.GRCh38.dna_rm.toplevel.fa
## AH49185    | Homo_sapiens.GRCh38.dna_sm.toplevel.fa
## AH49186    | Homo_sapiens.GRCh38.dna.toplevel.fa
## AH49187    | Homo_sapiens.GRCh38.ncrna.fa
## AH49188    | Homo_sapiens.GRCh38.pep.all.fa
```

We can use query to search the hub. The (possible) drawback to query is that it searches over different fields in the hub, so watch out with using a search term which is very non-specific. The query is a regular expression, which by default is case-insensitive. Here we locate all datasets on the **H3K4me3** histone modification (in H. sapiens because we selected this species above)

```
query(ah, "H3K4me3")
```

```
## AnnotationHub with 2018 records
## # snapshotDate(): 2015-08-17
## # $dataProvider: BroadInstitute, UCSC
## # $species: Homo sapiens
## # $rdataclass: GRanges, BigwigFile
## # additional mcols(): taxonomyid, genome, description, tags,
## #   sourceurl, sourcetype
## # retrieve records with, e.g., 'object[["AH23256"]]'
##
##           title
## AH23256 | wgEncodeBroadHistoneGm12878H3k4me3StdPk.broadPeak.gz
## AH23273 | wgEncodeBroadHistoneH1hesch3k4me3StdPk.broadPeak.gz
## AH23297 | wgEncodeBroadHistoneHeLas3H3k4me3StdPk.broadPeak.gz
## AH23311 | wgEncodeBroadHistoneHepg2H3k4me3StdPk.broadPeak.gz
## AH23324 | wgEncodeBroadHistoneHmecH3k4me3StdPk.broadPeak.gz
## ...     ...
## AH46826 | UW.Fetal_Muscle_Leg.H3K4me3.H-24644.Histone.DS21536.gappedPe
## AH46833 | UW.Fetal_Muscle_Trunk.H3K4me3.H-24851.Histone.DS23302.gappedPe
## AH46839 | UW.Fetal_Placenta.H3K4me3.H-24996.Histone.DS23300.gappedPe
## AH46845 | UW.Fetal_Stomach.H3K4me3.H-24639.Histone.DS22598.gappedPe
## AH46851 | UW.Fetal_Thymus.H3K4me3.H-24644.Histone.DS21539.gappedPe
```

Another way of searching a hub is by using a browser. Notice how we assign the output of `display` to make sure that we can capture our selection in the browser

```
hist <- display(ah)
```

RStudio

Project: (None)

Environment History

Files Plots Packages Help Viewer

Selected rows:

Return rows to R session

Show 10 entries Search: H3K4me3

idx	dataprovider	species	genome	description	tags	rdataclass	sourcetype
idx	dataprovider	species	genome	description	tags	rdataclass	sourcetype
AH40181	BroadInstitute	Homo sapiens	hg19	Bigwig File containing -log10(p-value) signal tracks from EpigenomeRoadMap Project	EpigenomeRoadMap, signal, consolidatedImputed, H3K4me3, E001, ESC, ESC.I3, ES-I3 Cells	BigWigFile	BigWig
AH40182	BroadInstitute	Homo sapiens	hg19	Bigwig File containing -log10(p-value) signal tracks from EpigenomeRoadMap Project	EpigenomeRoadMap, signal, consolidatedImputed, H3K4me3, E002, ESC, ESC.WA7, ES-WA7 Cells	BigWigFile	BigWig
AH40183	BroadInstitute	Homo sapiens	hg19	Bigwig File containing -log10(p-value) signal tracks	EpigenomeRoadMap, signal, consolidatedImputed, H3K4me3, E003, ESC, ESC.H1, H1 Cells	BigWigFile	BigWig

SessionInfo

```
## R version 3.2.1 (2015-06-18)
## Platform: x86_64-apple-darwin13.4.0 (64-bit)
## Running under: OS X 10.10.5 (Yosemite)
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## attached base packages:
## [1] methods      stats      graphics  grDevices  utils      datasets  base
##
## other attached packages:
## [1] AnnotationHub_2.0.3 BiocStyle_1.6.0      rmarkdown_0.7
##
## loaded via a namespace (and not attached):
## [1] Rcpp_0.12.0              AnnotationDbi_1.30.1
## [3] knitr_1.11              magrittr_1.5
## [5] BiocGenerics_0.14.0     IRanges_2.2.7
## [7] xtable_1.7-4            R6_2.1.0
## [9] stringr_1.0.0           httr_1.0.0
## [11] GenomeInfoDb_1.4.2     tools_3.2.1
## [13] parallel_3.2.1         Biobase_2.28.0
## [15] DBI_0.3.1              htmltools_0.2.6
## [17] yaml_2.1.13            digest_0.6.8
## [19] interactiveDisplayBase_1.6.0 shiny_0.12.2
## [21] formatR_1.2            S4Vectors_0.6.3
## [23] curl_0.9.2             mime_0.3
## [25] evaluate_0.7.2         RSQLite_1.0.0
## [27] stringi_0.5-5          BiocInstaller_1.18.4
## [29] stats4_3.2.1           httpuv_1.3.3
```

References

Hansen, Kasper D, Sarven Sabuncuyan, Ben Langmead, Noemi Nagy, Rebecca Curley, Georg Klein, Eva Klein, Daniel Salamon, and Andrew P Feinberg. 2014. "Large-scale hypomethylated blocks associated with Epstein-Barr virus-induced B-cell immortalization." *Genome Research* 24 (2): 177–84. [doi:10.1101/gr.157743.113](https://doi.org/10.1101/gr.157743.113).